

# Implementation Plan

## 1 Project Development Strategy

- Development will follow an **incremental integration approach** rather than attempting to build the complete system simultaneously.
- Each major subsystem will first be implemented and tested independently.
- Once a subsystem satisfies its individual requirements, it will be integrated with the other subsystems.
- The development process will progress from simulation and individual components toward a complete multi-robot demonstration.
- Each stage will produce measurable results that can be used to validate the following stage.

### Main development stages

1. Environment and simulation
2. Robot localization and SLAM
3. Autonomous exploration
4. Beacon communication network
5. Persistent spatial memory
6. Computer-vision victim detection
7. Wi-Fi CSI sensing
8. Multi-modal event detection
9. Outside Network integration
10. Executor navigation
11. Multi-robot coordination
12. Failure testing
13. Full-system validation

## 2 Environment and Simulation

### Objective

Create a controlled representation of the mine environment before testing the physical system.

### Implementation

- Model:
  - tunnel segments.
  - Intersections.
  - dead ends.
  - Entrances.
  - Obstacles.
  - areas of interest.
  - potential victim locations.
- Reproduce the GPS-denied nature of the Inside Area.
- Simulate multiple entrances so that multiple Writers can operate simultaneously.

- Include communication constraints between different parts of the environment.
- Define reference coordinates for validating the localization system.

### Validation

- Verify that the environment can be explored from unknown initial conditions.
- Verify that the simulated map can be compared with the ground-truth environment.
- Measure:
  - explored area.
  - exploration time.
  - localization error.
  - path length.

## 3 Robot Localization and SLAM

### Objective

Allow each Writer to estimate its own position and progressively construct a map without GPS.

### Implementation

- Integrate:
  - LiDAR.
  - IMU;
  - wheel encoders.
  - additional available sensors.
- Use sensor fusion to estimate robot motion.
- Implement SLAM to simultaneously:
  - estimate the robot pose.
  - construct the environment map.
- Maintain a local coordinate frame for each exploration mission.
- Associate detected events with the robot's estimated pose.

### Validation

- Compare estimated trajectories with known reference trajectories.
- Measure accumulated localization error.
- Test the system under:
  - straight corridors.
  - Turns.
  - Intersections.
  - Loops.
  - uneven surfaces.
  - temporary sensor degradation.

## 4 Autonomous Exploration

### Objective

Allow Writers to explore an unknown mine without requiring a predefined map.

### Implementation

- Detect exploration frontiers between known and unknown areas.
- Represent explored areas as a graph of connected regions.
- Use modified DFS to systematically explore tunnel branches.
- Calculate a utility value for available frontiers.
- Consider:
  - Distance.
  - estimated information gain.
  - Accessibility.
  - traversal cost.
  - mission relevance.
- Select the next exploration target automatically.
- Recalculate the exploration decision whenever new information changes the map.

### Validation

Test the algorithm on environments containing:

- simple corridors.
- multiple branches.
- dead ends.
- unequal branch lengths.
- blocked paths.
- multiple possible exploration frontiers.

### Measurements

- Percentage of environment explored.
- Exploration time.
- Total travelled distance.
- Number of revisited areas.
- Number of unexplored regions remaining.

## 5 Beacon Communication Network

### Objective

Create a reliable communication and persistent-information infrastructure inside the mine.

### Implementation

- Implement ESP-NOW communication between:
  - Writers and beacons.
  - neighbouring beacons.
  - Executors and beacons.
- Define a compact communication protocol.
- Each message should contain the information necessary for the receiving node to interpret and store the event.
- Implement beacon identifiers and robot identifiers.
- Implement message forwarding between neighboring beacons.
- Implement acknowledgement and retransmission mechanisms where required.
- Determine practical beacon spacing experimentally.

## Validation

Measure:

- communication range.
- packet-loss rate.
- Latency.
- successful forwarding rate.
- network recovery after a node becomes unavailable.

## 6 Persistent Spatial Memory

### Objective

Ensure that important information survives the failure or departure of the Writer that originally collected it.

### Implementation

Each relevant event should be represented by a structured record containing:

- Event ID.
- Event type.
- Position.
- Timestamp.
- Confidence.
- source robot.
- detection modality.

Examples of event types include:

- possible victim.
- confirmed victim.
- Obstacle.
- hazardous region.
- important environmental feature.

### Validation

- Store events on individual beacons.
- Forward them through multiple beacons.
- Remove or disable the original Writer.
- Verify that another robot can still retrieve the information.
- Verify that the information remains associated with the correct spatial location.

## 7 Computer-Vision Victim Detection

### Objective

Provide direct visual victim detection.

### Implementation

- Install and configure the camera system on the appropriate Writer.
- Collect representative images from the target environment.
- Develop or adapt the victim-detection model.

- Produce:
  - Detection.
  - Confidence.
  - image location.
- Associate each detection with the Writer's SLAM pose.
- Convert the observation into an event record.

### Validation

Test under:

- different lighting conditions.
- partial occlusion.
- different victim orientations.
- different distances.
- tunnel obstacles.

The output should be evaluated using detection accuracy and false-positive/false-negative rates.

## 8 Wi-Fi CSI Victim-Presence Detection

### Objective

Investigate whether Wi-Fi CSI can provide a reliable **Level-1 human-presence detection** capability inside the mine.

### Infrastructure

- Deploy Wi-Fi sources within the Inside Area using Writers and/or beacons.
- Deploy multiple ESP32-S3 CSI sensing nodes around the monitored region.
- Establish the CSI data aggregation and processing system.
- Maintain a clear separation between:
  - Wi-Fi CSI sensing.
  - ESP-NOW communication.

### Implementation

1. Configure the Wi-Fi sources.
2. Capture CSI measurements from the sensing nodes.
3. Establish a baseline for the environment without a person.
4. Record CSI changes caused by human presence.
5. Process and extract relevant signal features.
6. Train/test the AI classifier.
7. Produce a human-presence estimate.
8. Associate the detection with the relevant spatial region.

### Validation

Progressively test:

**Open environment → enclosed room → tunnel → obstructed tunnel → representative mine environment**

Measure:

- detection rate.

- false-positive rate.
- false-negative rate.
- detection latency.
- sensitivity to node placement.

### Phase 2 extension

- Increase spatial resolution using multiple CSI links.
- Investigate RF-based victim localization.
- Investigate fusion between RF localization, computer vision and SLAM.

## 9 Multi-Modal Event Detection

### Objective

Combine information from different sensing systems without making one sensing modality the sole source of truth.

### Implementation

- Receive observations from:
  - computer vision.
  - CSI.
  - other robot sensors.
- Associate observations with:
  - Position.
  - Time.
  - Confidence.
  - source.
- Determine whether multiple observations correspond to the same event.
- Generate a unified event record.

### Validation

- Test each sensing modality independently.
- Test them together.
- Measure whether combining observations reduces false detections.

## 10 Frame Translation

### Objective

Transform the Writer's locally estimated position into the global coordinate system required by the mission.

### Implementation

- Establish a local SLAM coordinate frame when the Writer enters the mine.
- Record the relationship between the entrance/reference beacon and the global coordinate system.
- Store the Writer's local pose in the beacon infrastructure.
- Use the known beacon reference to transform the local position into global coordinates.
- Associate detected events with the resulting global position.

### Validation

- Place reference points with known global coordinates.
- Compare calculated global positions with reference positions.
- Measure transformation error at different distances from the entrance.
- Repeat the test using different exploration paths.

## 11 Outside Network

### Objective

Provide the communication and energy interface between the Inside Area and the external infrastructure.

### Implementation

Each station can contain:

- communication computer.
- directional antennas/parabolic antennas.
- solar panels.
- high-capacity battery.
- robot charging system.
- optical-fibre connection.

The optical-fibre connection links the station to the first beacon of the corresponding entrance.

### Functions

- Receive mission information from the Inside Area.
- Forward information to the Command Post.
- Receive commands and mission information from the Command Post.
- Forward relevant information toward the entrance.
- Provide energy for robot recharging.

## 12 Command Post and Global Mission Database

### Objective

Allow information from previous missions to improve future missions.

### Implementation

- Command Post receives mission information from Outside Network stations.
- Store:
  - mission results.
  - detected events.
  - Maps.
  - Failures.
  - robot performance.
  - relevant environmental information.
- Share mission information with the global server.
- Allow information from previous missions to be consulted during future mission preparation.
- Maintain separation between:
  - local mission control.
  - global knowledge sharing.

## 13 Executor Robot

### Objective

Use persistent information generated by previous Writers to perform the required mission.

### Implementation

- Receive mission briefing from the Command Post.
- Obtain:
  - target information.
  - relevant position.
  - beacon information.
  - relevant map information.
- Enter the mine.
- Establish communication with the beacon network.
- Use beacon information for localization and navigation.
- Navigate toward the target.
- Execute the assigned task.

### Validation

- Create a target using information generated by a Writer.
- Remove the original Writer.
- Deploy the Executor.
- Verify that the Executor can reconstruct the route using the preserved information.

## 14 Multi-Robot Integration

### Objective

Demonstrate cooperation between different Writer and Executor types.

### Implementation

- Deploy multiple Writers through different entrances.
- Allow each Writer to explore an assigned region.
- Share relevant information through the beacon infrastructure.
- Allow Fast Writers to prioritize rapid event detection.
- Allow Slow Writers to prioritize systematic mapping.
- Allow the Command Post to dispatch Executors using accumulated information.

### Validation

Compare:

- single Writer.
- multiple Writers.
- Fast + Slow Writer configuration.

Measure:

- exploration time.
- area explored.

- number of detected events.
- information successfully transferred.
- Executor mission completion.

## 15 Full-System Validation

The final validation should reproduce the complete mission sequence:

1. Identify entrances.
2. Deploy Writers.
3. Explore unknown environment.
4. Construct SLAM maps.
5. Deploy beacons.
6. Detect events.
7. Store events in the beacon network.
8. Transfer information to the Outside Network.
9. Prepare an Executor mission.
10. Deploy Executor.
11. Navigate using persistent information.
12. Complete the mission.
13. Introduce controlled failures.
14. Verify system resilience.